FUNDAMENTAL OF DATA SCIENCE

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|| nd year CSE

Roll no:106

EXP : 1a Analyze the trend of data science job over the last decade

DATE:

Aim:

Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather

data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {'Year': list(range(2010, 2021)),

'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}


df = pd.DataFrame(data)

plt.plot(df['Year'], df['Job Postings'], marker='o')

plt.title('Trend of Data Science Job Postings')

plt.xlabel('Year')

plt.ylabel('Number ofJob Postings')

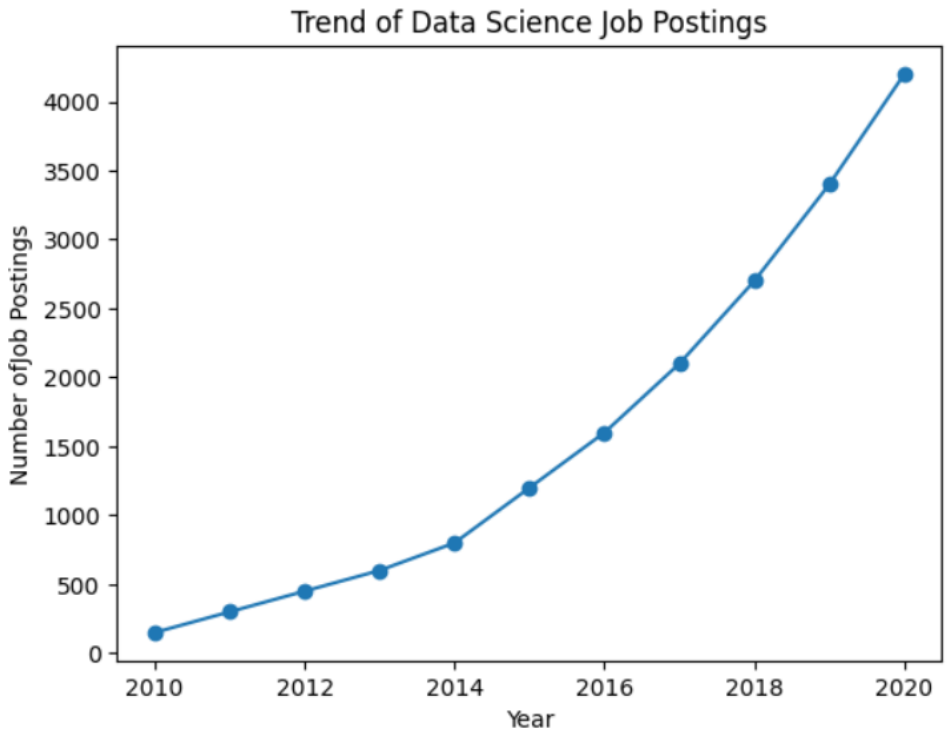
plt.show()
```

Year =2010, 2021

Job Postings=150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200

Output:

...



Result:

The above coding is executed successfully.

Exp No:1.b Analyze and visualize the distribution of various data science roles (DataAnalyst, Data Engineer, Data Scientist, etc.) from a dataset.

Aim:

Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

Code:

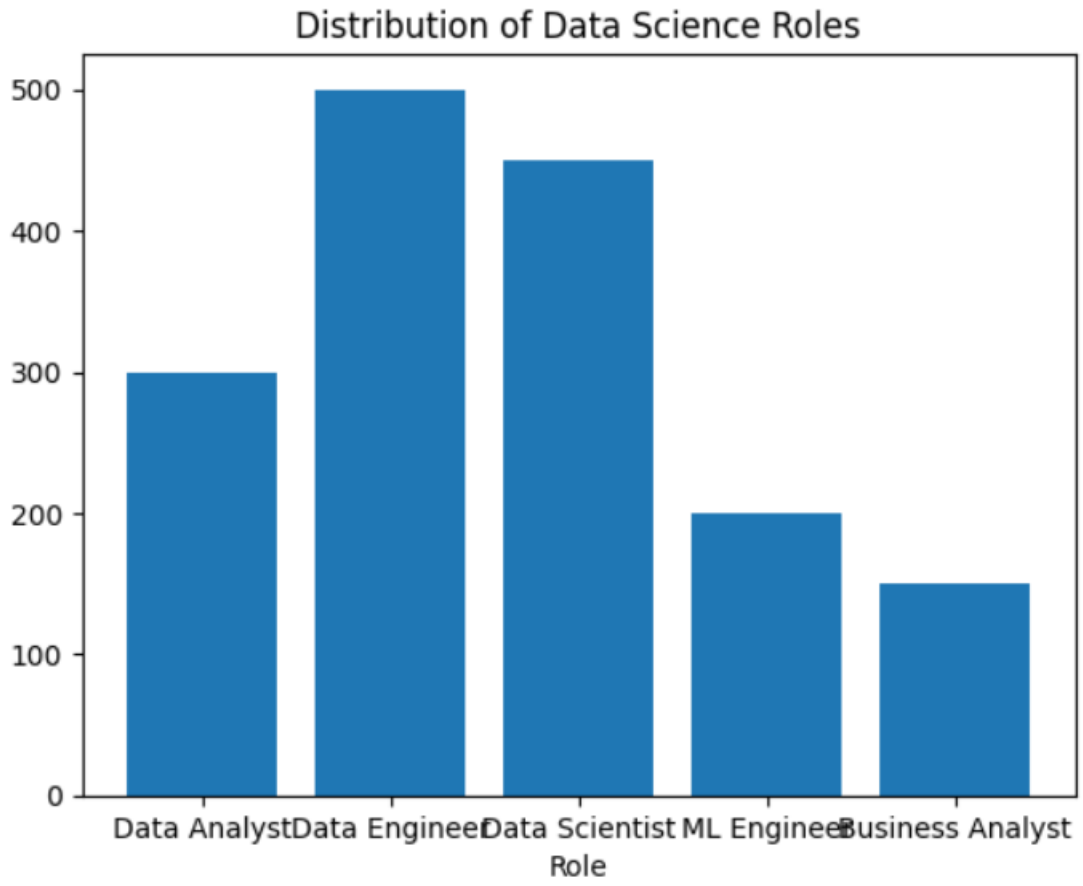
```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```

Sample Input:

roles = Data Analyst, Data Engineer, Data Scientist, ML Engineer, Business Analyst.

counts = 300, 500, 450, 200, 150.

Output:



Result:

The above code is executed successfully.

Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semistructured data based on data sets given.

Aim:

Create small datasets for each type and explain their characteristics.

Code:

```
structured_data = pd.DataFrame({
'ID': [1, 2, 3],
'Name': ['Alice', 'Bob', 'Charlie'],
'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)

unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image,
or a video file."
print("\nUnstructured Data:\n", unstructured_data)

semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

Output:

```
Structured Data:
    ID     Name  Age
0   1    Alice   25
1   2      Bob   30
2   3  Charlie   35

Unstructured Data:
 This is an example of unstructured data. It can be a piece of text, an image,
or a video file.

Semi-structured Data:
 {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
```

Result:

The above coding id executed successfully

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.

Aim:

Use the cryptography library to encrypt and decrypt a piece of data.

Code:

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()

f = Fernet(key)

token = f.encrypt(b"Rajalakshmi Engineering College")

token

b'...'

f.decrypt(token)

b'Rajalakshmi Engineering College'

key = Fernet.generate_key()

cipher_suite = Fernet(key)

plain_text = b"Rajalakshmi Engineering College."

cipher_text = cipher_suite.encrypt(plain_text)

# Decrypt data

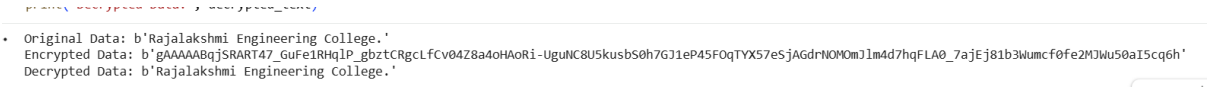
decrypted_text = cipher_suite.decrypt(cipher_text)

print("Original Data:", plain_text)

print("Encrypted Data:", cipher_text)

print("Decrypted Data:", decrypted_text)
```

Output:



Result:

The above coding is executed successfully